percentile. Relative to the baseline strategy, spending with this endowment strategy is 67 percent higher at the ninetieth percentile, 1 percent higher at the median, and 24 percent less at the tenth percentile. This strategy reduces downside volatility relative to the fixed-percentage strategy, as it also has a component of spending that uses constant inflation-adjusted spending. Across the board, average downside reductions to spending averaged 2 percent over the thirty-year period.

Next, the second endowment formula is based on a three-year moving average. The initial spending rate of 9.42 percent is just slightly below the initial spending rate with the fixed-percentage withdrawals. In practical terms, this strategy behaves similarly to fixed-percentage withdrawals but with slightly less volatility. In all cases, spending is reduced throughout retirement due to the high initial spending level. It is 63 percent less at the ninetieth percentile ranging to 86 percent less at the tenth percentile. In addition, because of the significant spending reductions over time across the distribution of spending, spending will be less by year thirty of retirement compared to the baseline constant inflation-adjusted spending strategy. The strategy has significant volatility on the downside as well, averaging 3 percent reductions in spending on a year-by-year basis over the thirty-year retirement period across the distribution of outcomes.

The next strategy is Bengen's dollar floor-and-ceiling withdrawal strategy. This strategy is based on allowing a ceiling that is 20 percent higher than the initial withdrawal rate level and a floor that is 15 percent less than the initial withdrawal level in dollar terms. The initial spending rate is 4.84 percent. Spending with this strategy is held within the range mentioned. It can reach as high as $5,810, which it does at the ninetieth percentile at subsequent points in retirement. This represents a 20 percent increase over initial spending. On the downside, spending can drop as low as $4,110, which is 15 percent below the initial retirement date spending level. By year thirty, spending has fallen to this floor at both the median and tenth percentiles. It is interesting that downside spending is only slightly less than with the constant inflation-adjusted baseline.

Bengen's dollar floor-and-ceiling rule allows initial spending to increase while also providing the ability to adjust spending in a way that reduces the